GOLDMAN SACHS

Summary Insights Report

Project: Loan Portfolio Analysis, Default Prediction and Business Recommendation

The SQL analysis identified the following distribution:

Loan Status	Number of Records	Percentage
Current	170,461	63.06%
Fully Paid	76,361	28.25%
Charged Off	17,851	6.60%
Other statuses	5,626	2.08%
Total	270,299	100.00%

Current loans represent the largest category in the dataset. Since these loans may not have reached a final repayment outcome, they should be monitored separately from Fully Paid and Charged Off loans.

The 17,851 Charged Off records indicate that historical charge-offs are an important area for further investigation. However, the proportion of charged-off records among all loans should not automatically be treated as the final default rate for the completed-loan portfolio.

2.2 Regional Customer Analysis

The regional analysis produced the following customer counts:

Region	Customer Records
South	97,683
West	69,137

Northeast 55,014

Midwest 48,465

The South has the largest recorded customer count among the regions shown. This information can help the bank understand customer distribution and plan regional reporting.

A higher customer count does not necessarily mean a higher default risk. Regional risk should be assessed using actual repayment outcomes and comparable default rates.

2.3 Purpose of SQL Analysis

SQL provides a structured way to inspect data, calculate counts, group loans by status, and compare customer distribution across regions. These summaries provide the foundation for the Power BI dashboard and the Python modelling work.

The SQL results should be used consistently across the project so that the figures shown in the dashboard and report agree with the underlying queries.

3. Power BI Dashboard Insights

The Power BI component presents loan portfolio information in a visual format to make trends and comparisons easier to understand.

The dashboard supports the review of loan status and portfolio composition. It can help users examine how the records are distributed across available categories and identify areas that require additional investigation.

One important finding from the underlying data is that Current loans form the largest status group. Therefore, the bank should distinguish loans that are still active from loans with final repayment outcomes when interpreting portfolio performance.

Regional and loan-purpose comparisons should be interpreted alongside the number of records in each category. Counts show the size of a group, whereas rates are needed to compare risk fairly.

For reliable reporting, each Power BI indicator should have a clear definition and should agree with the corresponding SQL calculation. The dashboard should not describe the Fully Paid share of all records as the repayment success rate among completed loans.

4. Python Machine Learning Analysis

4.1 Dataset Preparation

The Python analysis developed a Logistic Regression model to distinguish between two historical loan outcomes: Fully Paid and Charged Off.

The modelling dataset contained 94,212 records:

- Fully Paid: 76,361 records.
- Charged Off: 17,851 records.

The data was divided into training and testing sets using a stratified 80/20 split. Data preparation included handling missing values, encoding categorical variables, and scaling numerical variables.

The selected features included loan amount, loan term, issue year, annual income, customer account-balance information, loan purpose, home ownership, verification status, employment length, and state.

4.2 Model Evaluation

The Logistic Regression model achieved the following results on the test dataset:

Evaluation Metric	Result
Accuracy	63.55%
Precision for Charged Off	28.19%
Recall for Charged Off	59.72%
F1-score	38.30%

Accuracy measures the proportion of all test records classified correctly. Precision indicates how many loans predicted as Charged Off were actually Charged Off. Recall indicates how many actual Charged Off loans were identified by the model.

The model achieved 59.72% recall, meaning it identified approximately six out of ten charged-off loans in the test dataset. However, its precision of 28.19% shows that many loans flagged as charged off were actually fully paid.

4.3 Confusion Matrix Findings

The confusion matrix produced the following results:

Actual Outcome	Predicted Fully Paid	Predicted Charged Off
Fully Paid	9,842	5,431
Charged Off	1,438	2,132

The model correctly identified 9,842 Fully Paid loans and 2,132 Charged Off loans. It incorrectly flagged 5,431 Fully Paid loans as Charged Off and failed to identify 1,438 actual Charged Off loans.

These results show that the model can identify a portion of loans associated with charge-offs, but its false-positive count is substantial. The model should therefore support further review rather than automatically determine lending outcomes.

4.4 Feature Importance

The Logistic Regression coefficient analysis identified associations between default predictions and categories related to loan purpose, state, and home ownership.

Some categories had positive coefficients, while others had negative coefficients relative to their model reference categories. These results describe relationships learned from the historical dataset and do not prove that any factor causes default.

The bank should investigate these patterns further, consider the number of records in each category, and validate findings before changing lending policies.

5. Integrated Insights from SQL, Power BI and Python

The three project components contribute different types of insight.

SQL provides the underlying summaries of loan statuses and regional customer distribution.

Power BI presents portfolio indicators visually, helping users review the distribution of records and compare available categories.

Python adds a predictive analysis by estimating whether a loan belongs to the Charged Off or Fully Paid category, based on the selected historical features.

Together, these components demonstrate a workflow in which data is inspected and summarised, results are visualised, and machine learning is used to investigate potential risk patterns.

The findings should be interpreted together. Portfolio counts describe the size of groups, dashboard indicators help users explore patterns, and model predictions provide estimates rather than guaranteed outcomes.

6. Recommendations to Reduce Loan Defaults

1. **Monitor active loans: Track Current loans separately and use available repayment histories to identify emerging payment difficulties.**
2. **Use risk screening carefully: Use the Logistic Regression model to flag loans for additional review, not to automatically reject customers.**
3. **Investigate loan segments: Compare actual repayment outcomes across loan purposes, terms, amounts, and regions using clearly defined rates and adequate sample sizes.**
4. **Review financial information: Consider verified income, existing financial obligations, and repayment capacity as part of a responsible lending assessment.**
5. **Monitor model performance: Review precision, recall, false positives, and missed defaults as new repayment outcomes become available.**
6. **Maintain consistent reporting: Ensure SQL calculations and Power BI indicators use the same definitions and denominators.**

7. Approval Policy and Data Limitations

The available dataset does not contain genuine Approved and Rejected loan application labels. Consequently, a reliable loan approval prediction model could not be trained using the available data alone.

Fully Paid and Charged Off are repayment outcomes after a loan has been issued. They cannot be treated as substitutes for application approval decisions.

To develop an approval prediction model, Goldman Sachs would need historical application records containing genuine approval and rejection decisions and relevant applicant information.

The dataset also does not contain a genuine credit-score field. Loan grades, where available, should not be described as actual credit scores.

Any lending recommendations should be based on verified information, transparent criteria, appropriate human review, and applicable fairness requirements.

8. Recommendations for Portfolio Health

- **Monitor Current, Fully Paid, Charged Off, and other loan statuses separately.**
- **Define default rates and repayment indicators clearly, including their denominators.**
- **Compare regional and loan-purpose performance using actual repayment outcomes rather than record counts alone.**
- **Improve data collection for application decisions, repayment history, and relevant financial indicators.**
- **Validate Power BI measures against SQL results.**
- **Review model performance regularly and treat predictions as supporting evidence rather than guaranteed outcomes.**

9. Conclusion

The Goldman Sachs loan analysis project combines SQL, Power BI, and Python to examine historical loan records, visualise portfolio information, and investigate potential default risk.

The SQL analysis identified the distribution of loan statuses and regional customer counts. Power BI provides a visual way to review portfolio indicators, while Python demonstrated how Logistic Regression can identify loans associated with charge-off outcomes.

The model achieved 63.55% accuracy and 59.72% recall, but its precision of 28.19% indicates that many loans flagged as charged off were actually fully paid. The model therefore requires careful human review before any operational use.

The key priorities are to monitor repayment outcomes consistently, validate dashboard indicators, investigate risk patterns using reliable rates, improve data quality, and obtain genuine application decision labels before developing an approval prediction model.